ZPDPatch: Trading Repair Coverage for Student Code in Programming-Exercise Feedback

***Abstract*—We present ZPDPATCH, a trajectory-supervised technique that aims repair feedback at a student's own code by constructing targets from execution evidence rather than from accepted solutions. A repair returned to a student is feedback, so how much of their program it replaces matters alongside whether the tests pass. We call repair whose supervision depends on that history *trajectory-conditioned*, and no single comparison establishes a benefit for it, because more candidates alone improve a history-aware system and a new target rule also changes the training population. Our goal is to separate the target rule's effect from those of extra candidates and serialized history. ZPDPATCH mines three execution-evidence relations, trains independent adapters, and selects a validation-frozen portfolio behind a gate that returns the submission unchanged rather than a regression. On 181 student-held-out Introductory Programming (CS1) Java trajectories under source- and count-matched training, terminal targets repair 155 and progress targets 92, leading on 15 of 17 exercises. On the 90 trajectories both rules repair, the retention advantage is zero for 63.3% of students, positive for 31.1%, and negative for 5.6%. An online-judge corpus of 997 inputs from problems seen in training and 250 from held-out problems reproduces the tradeoff, with 19.2 and 11.2 percentage points less coverage and 5.6 and 6.5 points more retained tokens. Three fixed candidate draws alone add 15.1 points of repair rate, and these numbers describe the feedback, not learning. We therefore retain the current-only terminal portfolio.**

***Index Terms*—programming-exercise feedback, automated program repair, submission trajectories, target construction, CS1**

I. INTRODUCTION

Automated Program Repair (APR) follows a generate-and-validate loop, generating changes to a buggy program and keeping one that passes the tests [1]–[4]. In a programming course such a repair is feedback, and the student receives a patch personalized to their own submission rather than only an autograder's verdict [5]–[7]. To use that feedback the student has to find what the returned program changed in their own, and it helps only as far as they can locate and understand that change [8], [9].

Automated repair for programming assignments returns a program that passes every test, built by clustering, aligning, or normalizing correct peer programs [10]–[12], by generating and diagnosing repairs with language models [6], [7], [13]–[15], or by retrieving a learner-specific solution to steer generation [16]. Because this line is supervised toward an accepted solution, what the student gets back is an answer rather than a step, and the returns of the strongest retrieval-based baseline sit 88.27 AST edits from the submission on average (Section IV). A second line returns a next-step hint instead. The student receives one edit, as a short message or a highlighted location in the editor, taken from a path toward a matched correct program in a space built from past correct solutions [17]–[20]. Two submissions at the same place in that space therefore receive the same edit, whatever tests each of them currently passes, and because a described edit is not a program, what it would pass and how much it would replace are not in the return. Between them the two lines have made automated feedback practical in courses, raising how often an incorrect submission becomes a passing program and putting a next step within reach of far more students than a human tutor can see [5], [21].

Neither line is evaluated under a bound on how much of the student's program its return may replace. The Zone of Proximal Development motivates fitting support to the step a learner can take next [22]–[25], and we narrow that idea to one measurable requirement, that the returned program keep more of the submission. A target that can meet it is the student's own next improvement rather than an accepted solution, which raises whether that target must come from this student's history at all. We define *trajectory-conditioned repair* as repair whose supervision or generated change depends on the sequence leading to the current submission. The closest prior work, LSGen, also mines chronological wrong-to-later-correct pairs, but it uses them for retrieval rather than as supervision and it does not separate what the history contributes from what extra candidates contribute [16]. The Answer rule emits 40,454 examples and the Progress rule 21,416, so the training population moves with the target, and three fixed stochastic draws alone add 15.1 points of repair rate (Section II-C). A gain measured without holding sources, example counts, and candidate budgets fixed therefore cannot be attributed to the target. What we measure is what the returned program keeps, not what a student learns from it, since a smaller patch does not establish learning.

We present ZPDPATCH, a Mine–Train–Select technique that builds repair targets from a student's own submission sequence. *Mine* pairs an incorrect submission, or a retained prefix, with a later submission by three rules. Answer takes the accepted submission the trajectory reached, Strict the next better-verdict state, and Progress the next state that improves the verdict or passes more tests at an unchanged verdict. *Train* fits an independent adapter per rule, and *Select* freezes a validation-selected checkpoint triple. An execution gate returns a repair or retains the buggy program, optionally within an edit budget, so a course can bound how much of a submission a repair may replace and never receive one that lowers its

observed pass rate. Source-matched retraining, same-target independent seeds, matched budgets, and a matched-peer target control separate the target's contribution from the sources and the budget.

Under a bound on how much of a submission a return may replace, our target rule leads the strongest prior method at every bound but the broadest. At an edit budget of 20 AST operations it repairs 44.0% of 997 inputs from problems seen in training where LSGen repairs 13.9%, and across six prespecified budgets its mean advantage over LSGen is 14.7 points. LSGen passes us only once the bound is loosened to 160. Matched Answer targets repair 19.2 and 11.2 points more of those and of 250 inputs from held-out problems, Progress targets increase source-token retention by 5.6 and 6.5 percentage points where both rules repair, and on a further 9.4 points of the former the Progress rule returns a program that passes more tests without passing all. Student-disjoint CS1 Java retraining reproduces both directions, and the tradeoff is uneven, with Answer targets leading on 15 of 17 exercises and the retention advantage reaching 31.1% of the students both rules repair. Our portfolio over the three rules reaches 61.5%/74.8% unrestricted, below the 72.0%/81.2% of an Answer triple that sees only the current submission and below LSGen's 80.2% on the former. We retain the current-only Answer triple, and report the Progress rule as the option a course takes when locality binds.

Contributions. Our contributions are the following three.

- **Observation.** Prior work is evaluated on whether a return passes the tests, not on how little of the student's program it replaces. We define repair that targets that submission's next improvement as trajectory-conditioned repair, and show that attributing a gain to that target requires holding sources, example counts, and candidate budgets fixed.
- **Technique.** We specify Mine, Train, and Select, including the three pairing rules, portfolio selection on executed validation repairs rather than training loss, and an execution gate with explicit fallback and edit-budget semantics.
- **Evaluation.** On Python and student-disjoint CS1 Java we measure that target construction moves coverage and locality in opposite directions, separate the effects of the target, its source, the decoder, and the checkpoint, resolve that tradeoff across exercises and students instead of reporting cohort means, and map every claim to a retained artifact.

II. Preliminaries

A. Submissions and Execution Evidence

For a problem p, a submission state is a tuple $s = (c, v, o)$, where c represents complete source code, v denotes the recorded online-judge verdict, and o maps each available testcase identifier to its re-executed outcome. The problem context d_p contains the statement and resource limits, and $\mathcal{E}_p$ is the available test suite. A trajectory $\tau = [s_1, \ldots, s_n]$ orders submissions by one user on one problem. The canonical dataset ends each trajectory at its first recorded acceptance; this endpoint is the final accepted submission within the retained trajectory, not a later post-acceptance revision. Recorded verdicts and replayed outcomes remain distinct fields, because runtime versions and execution variability can change a program's observed behavior.

Require: Current code c, context x, ordered generators S, tests $\mathcal{E}$, edit budget B
Ensure: Returned program c^*
$c^* \leftarrow c$; $q^* \leftarrow \textsc{Execute}(c, \mathcal{E})$
for $v \in S$ **do**
  $z \leftarrow \textsc{Generate}(v, x)$
  if $B < \infty$ and $\mathrm{ted}^\dagger(c, z) > B$ **then**
    continue
  $q \leftarrow \textsc{Execute}(z, \mathcal{E})$
  if $q = 1$ **then**
    return z
  if $(q, -\mathrm{ted}^\dagger(c, z)) > (q^*, -\mathrm{ted}^\dagger(c, c^*))$ **then**
    $c^* \leftarrow z$; $q^* \leftarrow q$
return c^*

Fig. 1. Execution-gated repair procedure.

A training example pairs input $x = (d_p, h_i)$ with later code y, where h_i is the permitted submission prefix ending at the current state. A target constructor determines both the retained prefix and its target. At inference, the current program is always available; a history-conditioned prompt also includes earlier observed submissions, whereas a current-only prompt does not. The future accepted program is a training target or an offline assessment reference, never part of the inference prompt. A *terminal target* is the Accepted (AC) endpoint. A *progress target* is the next retained state under an execution-evidence rule and can itself be accepted. We reserve *intermediate target* for a retained nonterminal state; the Progress dataset is not restricted to such states.

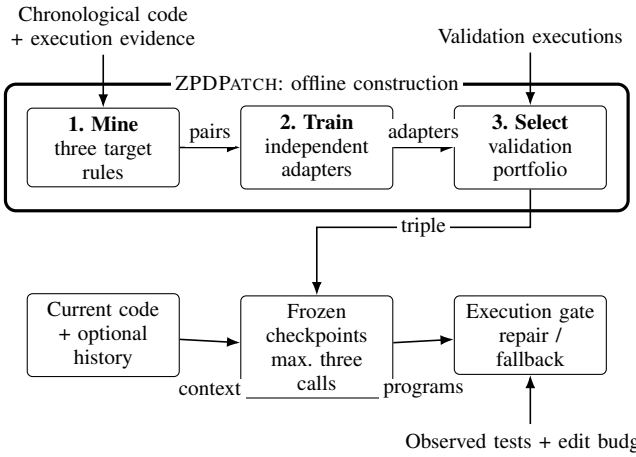
B. Generate-and-Validate Selection

Figure 1 specifies the common controller: the buggy program is the fallback, each generator emits one complete candidate that a finite edit budget can exclude, and execution either accepts a candidate immediately or keeps the best partial repair. Here q is the fraction of tests passed, and $\mathrm{ted}^\dagger$ is Abstract Syntax Tree (AST) edit distance when both changed programs parse, zero for unchanged code, and infinity otherwise. An unrestricted budget bypasses the distance check, allowing syntax repair; a finite budget rejects changes whose source distance is undefined.

The controller separates generator construction, input context, and opportunity count. Reference-based, learned, and iterative repair instantiate these choices differently [10], [14], [26]. Our controls compare specific implementations, not an assumption that all prior methods train on accepted endpoints.

C. Limitation and Goal

Consider three wrong-answer submissions with pass vectors $(1, 0, 0)$, $(1, 1, 0)$, and $(0, 1, 1)$, followed by an accepted state. The second preserves every previously passed test and gains



(a) stages and deployment

Require: Trajectories H, evidence O, validation inputs U, seeds Σ, portfolio size k, edit budget B
Ensure: Frozen ordered checkpoint portfolio S

$(D_A, D_S, D_P) \leftarrow (\emptyset, \emptyset, \emptyset)$ ▷ Mine
for $\tau = [s_1, \ldots, s_n] \in H$ **do**
  $D_A \leftarrow D_A \cup \mathsf{TerminalPairs}(\tau)$
  for $\rho \in \{S, P\}$ **do**
    $R \leftarrow [s_1]$
    for $s_t \in [s_2, \ldots, s_n]$ **do**
      if $\mathsf{Improves}_\rho(R[-1], s_t; O)$ **then**
        $D_\rho \leftarrow D_\rho \cup \{(R, c_t)\}$; $R \leftarrow R \,\|\, [s_t]$
$V \leftarrow \emptyset$ ▷ Train
for $(\rho, \sigma) \in \{A, S, P\} \times \Sigma$ **do**
  $V \leftarrow V \cup \{\mathsf{Train}(D_\rho, \sigma)\}$
for $v \in V$ **do** ▷ Select
  $R_v^B \leftarrow \mathsf{Validate}(v, U, B)$
$S \leftarrow \mathsf{Select}(\{R_v^B\}_{v \in V}, k)$
return $\mathsf{FreezeAndOrder}(S)$

(b) construction procedure

Fig. 2. ZPDPATCH. Panel (a) separates offline construction from online repair, and panel (b) returns the frozen portfolio whose checkpoints supply the generators of Figure 1.

TABLE I
CANONICAL TRAINING RULES BEFORE MATCHED CONTROLS. COUNTS ARE EXAMPLES, NOT UNIQUE SOURCES OR TEST SUCCESSES.

Rule	Target and context	Examples
Answer	Each non-AC source alone to the accepted endpoint	40,454
Strict	Retained prefix to the next better-verdict state	16,973
Progress	Retained prefix to the next verdict/test-progress state	21,416

another; the third gains a test but loses the first. Verdict-only filtering cannot distinguish them, while test-wise progress retains the second but not the third relative to it. Terminal supervision instead pairs each failed source independently with the acceptance. This example illustrates different training signals, not evidence that the intermediate change is smaller or easier for a student.

TABLE II
CORPUS AFTER WHOLE-TRAJECTORY FILTERING. PART.: PARTITION; SUBM.: SUBMISSIONS; PREFIXES PRECEDE RELATION FILTERING. SEEN PROBLEMS/TESTS ARE SHARED ACROSS ITS PARTITIONS; USER COUNTS ARE NOT ADDITIVE.

Group	Part.	Probl.	Tests	Traj.	Subm.	Prefixes	Users
Seen	Train	492	9,446	14,027	54,481	40,454	4,012
	Valid	492	9,446	1,754	6,719	4,965	1,248
	Test	492	9,446	1,754	6,598	4,844	1,250
Unseen	Test	54	642	259	954	695	236
Total		546	10,088	17,794	68,752	50,958	–

Table I shows why exchanging a target rule is not a one-variable intervention: example counts and source frequencies change together. The relation-level comparison characterizes complete methods, whereas matched-source retraining estimates target replacement on a common population. A scoped eight-study audit identifies five studies reporting multiple generation opportunities [7], [14]–[16], [21]. Their accounting differs: pass@1/5/10, conversation aggregation, and LSGen's 0–3 retrieval iterations do not estimate the same intervention. Our fixed-draw contrast holds generated candidates constant to isolate opportunity count. The artifact supplies the eight-study coding and sources; this purposive map is not a systematic review.

Our goal is to test trajectory-based supervision against current-code and budget-matched alternatives. Correctness, locality, and history specificity are separate outcomes. Later development additionally requires non-increasing regressions and unknowns under fixed controls.

III. OUR APPROACH

A. Overview

Our approach separates offline construction from online repair. Mine constructs datasets, Train fits adapters, and Select freezes a portfolio using executed validation repairs rather than training loss. Checkpoints generate independently from the same permitted input, without exchanging candidates. The mixed pool contains three seeds per relation; Answer-only and current-only variants are controls. The recommended current-only Answer-3Seed remains a comparator, not a renamed history model. The three stages run once per corpus, after which deployment serves repairs from the frozen checkpoints and issues at most one call per selected checkpoint, so the deployed budget is three calls.

B. Main Algorithm

The primary configuration uses three seeds per relation and $k = 3$, and breadth and pool controls vary these opportunities. Figure 2(a) shows those two phases, Figure 2(b) constructs the frozen checkpoint portfolio, and Figure 1 deploys it to repair an input.

C. Mine

The goal of Mine is to make every supervision pair traceable to an observed source and a later submission. Submissions are

grouped by problem and user, chronologically ordered, and cut at the first acceptance. Canonical normalization removes consecutive duplicates while preserving the source code used for generation and execution. The builder imposes no maximum history or transition count. A whole-trajectory filter instead excludes a trajectory before splitting if any required prompt–target configuration exceeds 4,096 tokens. Exclusion changes the evaluated population; it never cuts off a program or history and presents the fragment as complete.

The coarse severity function assigns 0 to Accepted (AC), 1 to Wrong Answer (WA), 2 to Time or Memory Limit Exceeded (TLE/MLE), 3 to Runtime Error (RE), and 4 to Compile Error (CE); unrecognized and internal outcomes occupy the most severe category. This is an operational ordering, not semantic distance to correctness. Strict compares recorded verdicts. Progress additionally compares replayed outcomes when the recorded verdict strings agree. Its cache must be complete, and compared vectors must have equal, nonempty testcase keys. Provenance includes problem, code, tests, and execution configuration. Infrastructure uncertainty in later development is reported as unknown, not converted into a favorable repair label.

For equal testcase universes, $o_t \prec_\times o_r$ holds when every test in o_t has severity no greater than its counterpart in o_r, with a strict decrease on at least one. The improvement predicates are:

$$\begin{aligned} \mathsf{Improves}_S(r,t) &\iff \mathrm{sev}(v_t) < \mathrm{sev}(v_r), \\ \mathsf{Improves}_P(r,t) &\iff \mathsf{Improves}_S(r,t) \\ &\quad \vee\ (v_t = v_r \wedge o_t \prec_\times o_r). \end{aligned} \tag{1}$$

Intuitively, Equation (1) retains coarse improvements and progress hidden by an unchanged verdict. A severity improvement such as RE to WA need not increase the pass fraction, and a verdict-improving edge does not require test-wise dominance. The mined relation supplies supervision, not a deployment-safety guarantee.

Answer emits an independent pair for each non-AC source and the accepted endpoint. Thus $[s_1, \ldots, s_9]$ yields $([s_1], c_9), \ldots, ([s_8], c_9)$; it does not expose accumulated prefixes to Answer training. Strict and Progress start with $R = [s_1]$, compare each later state with the *last retained* state, and append it only when the predicate holds. Before appending s_t, the builder emits (R, c_t). For WA, improved WA, AC, Strict emits $([s_1], c_3)$, whereas Progress emits $([s_1], c_2)$ and $([s_1, s_2], c_3)$. The accepted endpoint remains eligible in both rules. Although Strict's pairwise predicate is contained in Progress's, their independently scanned edge sets need not be nested.

Chronological indices strictly increase, so construction terminates without cycles. With fixed-key evidence, retained edges also decrease the appropriate verdict/test severity rank. These are builder invariants, not evidence of a learned ZPD effect. Binary accepted/non-accepted mining retains 14,027 of 16,973 Strict pairs and 18,009 of 21,416 Progress pairs. Models retrained on them agree with the canonical ones on 84.6% and 87.5% of Seen repair decisions (Section IV-E), so we read the ranking as operational rather than universal.

D. Train

The goal of Train is to learn each relation independently while keeping model and optimization comparable. The implementation fine-tunes Qwen2.5-Coder-7B-Instruct with 4-bit Quantized Low-Rank Adaptation (QLoRA) [27], [28]. Adapters have rank 16, scaling 32, and dropout 0.05, targeting attention and feed-forward projections. Canonical training uses one epoch, learning rate 2×10^{-4}, and seeds 2027–2029. Assistant target tokens supply cross-entropy loss. The role-neutral prompt contains the statement, limits, permitted submissions, execution/edit summaries, and a request for a complete solution. It names neither the rule nor a preferred patch size, so specialization must come from the data rather than explicit instructions.

Targets are future submissions available offline; inference inputs contain only observed states. Current-only controls reset positions and remove earlier entries, preventing length leakage through position labels. Answer training is already single-source, so supplying history at inference creates a distribution change. The history-removal retraining control instead holds targets fixed and changes earlier-submission serialization. Source-matched target training holds current input and count fixed. These designs answer different questions and are not pooled into one claim about history.

The paired-target control inner-joins records by problem, user, submission identifier, and the exact buggy program, retaining sources with distinct Answer and Progress targets. Its 7,389 training and 931 validation examples per condition use identical current-only inputs. The separate peer control requires another user on the same problem and in the same split, with exactly equal target execution evidence. Writing y_u for the same-user target and y_v for its peer replacement, for semantic-token diff distance d the discrepancy $\delta = |d(c, y_v) - d(c, y_u)|$ must satisfy $\delta \le 2$ or $\delta / \max(1, d(c, y_u)) \le 0.1$. Scarce evidence groups are visited first; ties prioritize distance discrepancy, reuse, and identifier, with donor reuse capped at three. This greedy rule yields 5,231 training and 306 validation examples per condition. Equal outcomes and nearby distances do not equate structure or style, so its measured effect belongs to the entire replacement rule, not isolated provenance.

E. Select

The goal of Select is to deploy a bounded portfolio chosen without final-test outcomes. For validation inputs U, let R_v^B be inputs repaired by checkpoint v within edit budget B. Selection maximizes:

$$f_B(S) = \frac{1}{|U|} \left| \bigcup_{v \in S} R_v^B \right|, \qquad |S| = 3. \tag{2}$$

Intuitively, Equation (2) rewards complementary successes, not low training loss or the presence of every relation. Validation uses one fixed-hash trajectory from each of 461 eligible problems. The mixed pool has nine checkpoints and 84 triples;

requiring one checkpoint per relation instead permits 27. The primary unconstrained mixed choice is Answer2027, Answer2028, and Progress2027. The relation-constrained choice is a different condition. Answer-9Choose3 receives nine Answer checkpoints and the same 84-triple search. Ties use recorded pass-rate, improvement, and deterministic identifier priorities. Budget-indexed selection repeats the declared objective at $B \in \{5, 10, 20, 40, 80, 160\}$, without choosing a budget from test results.

Selected checkpoints and order are frozen. Each emits at most one independent candidate, and the first observed acceptance stops generation. If none passes all tests, Figure 1 prioritizes pass fraction and then smaller distance. Because unchanged code is feasible, for the same fixed execution measurements the returned pass fraction cannot decrease, and a finite edit budget cannot be exceeded. This is not per-test non-regression: equal fractions can conceal gains and losses on different tests. It also does not guarantee non-regression under fresh execution, an unseen suite, or resource variability. Development explicitly measures these stronger failure modes.

For fixed independent candidates, order changes calls and patch identity but not the existence of an eligible acceptance. Stored candidates permit exact selection replay, not a guarantee of generation quality. Terminal pairing is linear in trajectory length; replaying n submissions on m tests requires $O(nm)$ executions and evidence storage. The 84-triple search is bounded at our pool size, not a scalable optimizer for arbitrary pools.

IV. Experiments

We evaluate the retained implementation through four research questions:

- **RQ1: Repair effectiveness.** Which configuration provides the strongest unrestricted repair performance?
- **RQ2: Target construction and locality.** How do matched targets affect correctness and locality, and how do both distribute across exercises and students?
- **RQ3: Component efficacy.** What do candidate breadth, checkpoint identity, and serialized history contribute?
- **RQ4: Generality.** Which findings replicate across problems, prefixes, scales, and Java populations?

A. Experimental Settings

Benchmarks. Table II describes the Project CodeNet Python800 corpus [29]. We retain trajectories with at least three submissions, a non-AC state before the first acceptance, and an accepted endpoint in the benchmark reference set. Users must appear in at least three problems, and retained problems contain at least two trajectories. The refined corpus has 546 problems, 21,454 trajectories, 99,432 submissions, and 10,088 tests. Whole-trajectory 4,096-token filtering removes 3,660 trajectories before splitting. Problems ranked by eligible trajectory count supply the top 90% to Seen (492) and the remaining 54 low-resource problems to Unseen. Seen trajectories are split 80:10:10 within every problem, retaining at least one in each partition.

The primary endpoint is the last recorded non-AC submission of an eventually successful trajectory. Runtime eligibility requires the buggy program to remain non-AC and the recorded accepted endpoint to pass the available tests on replay. This leaves 997 of 1,754 Seen test trajectories, from 328 problems, and 250 of 259 Unseen trajectories, from all 54 problems. Excluded trajectories do not enter the denominator. CodeNet users are online-judge participants, not a verified classroom population. We also use TIKTOC's release of authentic Java CodeWorkout submissions and tests from an Introductory Programming (CS1) course [30]: 1,125 trajectories from 224 students and 17 exercises, with 181 test trajectories from 33 held-out students. CodeWorkout is a practice platform on which a course assigns short method-level exercises with instructor-authored tests; students submit repeatedly and see which tests fail, the setting a returned repair would enter. This corpus supplies the classroom population and CodeNet the scale. A separate exercise split has 10/3/4 train/validation/test exercises and 605/216/304 trajectories. These populations remain separate from the Python estimates.

Baselines. Zero-shot uses the same base model without fine-tuning and appends verdict/pass-count feedback after failure. LSGen retains repair-pair filtering, edit-vector retrieval, diff-guided generation, and re-retrieval [16]; its same-problem Seen-Train references are unavailable under our Unseen information constraint, so that condition is not evaluated. Answer-1 uses one Answer checkpoint. Answer-3Seed uses seeds 2027–2029 on the same 40,454 Answer pairs, controlling independent training without heterogeneous targets. Answer-9Choose3 adds seeds 2030–2035 and selects three from nine. History and current-only prompts are explicitly labeled. Mixed-9Choose3 denotes the primary unconstrained ZPDPATCH pool. Current-only reselection from the mixed pool chooses the same three Answer checkpoints as Answer-3Seed, not a relation-mixing improvement.

Other settings. Training and generation use one NVIDIA RTX 5090. Canonical Python evaluation runs one subprocess per test with a 2.5-second timeout and a 2GB address-space limit, whereas Java 21 runs in a network-disabled, read-only container (16GB shared) with 15-second compilation and whole-harness limits; these are different harnesses, not a common per-test budget, and the manifest does not pin the Python minor version. The primary comparison permits three generations of at most 4,096 output tokens each. Historical Transformers and the later vLLM factorial remain separate stacks. Matched-target Python/Java training uses batch size 2, accumulation 8, one epoch, and minimum validation-loss checkpoint selection. Each target rule uses all three seeds in order 2027–2029, greedy decoding, current-only prompts, and the same controller; it does not search nine-checkpoint portfolios. The supplement maps these conditions and source hashes. Portfolio outcomes are not means over three independent test runs.

Repair Rate (RR) is the fraction of eligible inputs whose returned program passes every test. Pass Rate (PR) averages passed-test fractions; Improvement Rate (IR) records

an increased fraction. Tree Edit Distance (TED) between ASTs measures current-to-returned change [31], [32] and is undefined when either program does not parse. Undefined cases remain in unrestricted RR but not TED summaries; finite gates reject undefined-distance changes. Unpaired TED averages successful, parseable returns; paired locality uses joint successes. Retention divides tokens or nonempty stripped lines matched by `SequenceMatcher` (autojunk disabled) by the source count. Python tokenization omits layout tokens but keeps comments; Java separates words and punctuation. Retention differences are percentage points (pp), not token counts or learning measures.

Unless stated otherwise, brackets give 95% percentile bootstrap intervals from 10,000 problem-cluster resamples with replacement, preserving all paired inputs in each sampled cluster [33]. Java separately resamples students and exercises. Instance-weighted estimates divide pooled differences by sampled inputs; equal-problem estimates average problem means. Exact two-sided McNemar tests assume independent input pairs and are supplementary to cluster-aware intervals, not substitutes for them [34]. Holm adjustment covers the four history-removal retraining tests (Strict/Progress $\times$ Seen/Unseen), not every exploratory contrast [35]; individual-rate Wilson intervals are reported separately [36]. Protocols identify when controls were frozen, but the extended study was not externally preregistered. Adaptively inspected development remains exploratory. We retain whitespace-sensitive corrections and original counts separately.

Research process. OpenAI Codex assisted with design, code, orchestration, diagnostics, and manuscript preparation under author instructions, and automated agents executed authorized jobs; we do not claim authors manually ran every job. Outcomes come from retained execution records, not generated prose. The experiment program stopped at the declared cutoff of 21 October 2026, 13:00 KST. This synthesis adds no training, generation, testcase rerun, or model promotion.

B. RQ1: Repair Effectiveness

TABLE III
PRIMARY PYTHON EVALUATION. RR DIVIDES SUCCESSFUL RETURNS BY 997 SEEN OR 250 UNSEEN INPUTS; TED AVERAGES DEFINED DISTANCES AMONG SUCCESSFUL RETURNS ONLY. PORTFOLIOS PERMIT AT MOST THREE CALLS; ANSWER-1 PERMITS ONE. H: HISTORY INPUT; C: CURRENT-ONLY. NO CROSS-SPLIT RANKING IS IMPLIED.

Method	Input	Seen		Unseen	
		RR (%)	TED	RR (%)	TED
Zero-shot	H	30.7	27.25	62.0	17.46
LSGen	H	80.2	88.27	–	–
Answer-1	C	58.1	30.27	69.6	15.30
Answer-3Seed	H	61.7	24.52	73.2	13.38
Answer-9Choose3	H	59.8	25.46	74.8	15.19
Mixed-9Choose3	H	61.5	21.05	74.8	11.40
Answer-3Seed	C	72.0	37.94	81.2	16.01
Answer-9Choose3	C	69.5	35.41	82.0	20.06

Results. Table III identifies current-only Answer-3Seed as the retained deployment configuration within our implementation, at 72.0% Seen and 81.2% Unseen. It repairs 718/997 and 203/250 inputs. Relative to current-only Answer-1, its paired gain is 13.9 points [11.71, 16.29] and 11.6 points [7.32, 16.23]. This contrast changes both checkpoint identity and opportunity count; RQ3 separates them. It is not universally best: current-only Answer-9Choose3 repairs two more Unseen inputs (205/250), but 25 fewer Seen inputs (693/997). We retain the simpler fixed triple rather than selecting a different final method for each observed test split. Current-only mixed-pool reselection chooses exactly that Answer triple, providing no evidence that relation diversity improves this deployment setting.

The history-input mixed portfolio repairs 613/997 Seen and 187/250 Unseen inputs. Its Seen gain over Zero-shot is 30.8 points [26.43, 35.09], or 26.1 [21.24, 30.93] with equal problem weights. Mixed-only and Zero-shot-only successes number 342 and 35 (McNemar $p < 10^{-63}$). This is not a history-specific effect: Mixed nearly ties Answer-3Seed, while its matched nine-checkpoint contrast is +1.7 points [-0.79, 4.29] on Seen and 0.0 [-3.70, 3.75] on Unseen.

LSGen repairs 800/997 Seen inputs versus Mixed's 613/997, a difference of -18.8 points [-23.04, -14.46]; 80 versus 267 exclusive successes give $p < 10^{-23}$. On joint repairs, Mixed's TED is lower by 43.34 [38.47, 48.44] relative to LSGen and 8.46 [5.05, 12.55] relative to Zero-shot. Conditioning excludes unilateral successes, so smaller patches do not cancel the primary correctness deficit.

TABLE IV
SOURCE- AND COUNT-MATCHED TARGET REPLACEMENT. RR USES ALL ELIGIBLE INPUTS; RETENTION DIFFERENCES ARE PROGRESS MINUS ANSWER ON JOINT REPAIRS. JAVA USES HELD-OUT STUDENTS. POSITIVE RETENTION DIFFERENCES FAVOR LOCALITY, NOT DEMONSTRATED LEARNING.

Population	RR (%)		Retention (pp)	
	Answer	Progress	Tokens	Lines
Seen (997)	53.2	34.0	+5.6	+8.9
Unseen (250)	76.8	65.6	+6.5	+10.4
Java (181)	85.6	50.8	+4.8	+5.1

Answer to RQ1. We retain current-only Answer-3Seed, which is neither an overall winner nor a ZPD success: LSGen leads on Seen, Answer-9Choose3 is marginally higher on Unseen, and mixed targets have no established unrestricted advantage.

C. RQ2: Target Construction and Locality

Matched targets. Table IV changes the target while holding current inputs and training counts fixed. Progress-minus-Answer RR is -19.2 points [-22.96, -15.54] on Seen and -11.2 [-17.37, -4.88] on Unseen, with exact McNemar $p = 7.85 \times 10^{-34}$ and $p = 6.17 \times 10^{-5}$. Thus the canonical Answer dataset's advantage cannot be attributed only to having more training records. The paired population is restricted to sources with distinct targets, however, and does not estimate an effect for all CodeNet submissions. Moreover, a Progress

target can itself be accepted. The intervention is the retained-evidence target rule, not a clean contrast between exclusively incorrect and exclusively correct targets. A returned program can move a submission forward without finishing it. Under the matched inputs the Progress rule repairs 34.0% of Seen inputs and improves 43.4%, so a further 9.4 points receive a program that advances without passing, against 53.2% and 65.0% for the Answer rule. The same two gaps are 6.4 and 7.6 points on Unseen, and 9.4 against 2.2 on Java, where Answer targets already repair 85.6% outright. Part of the coverage the Progress rule gives up therefore returns as partial progress, counted by test execution and not by comprehension.

Progress's locality benefit is visible on joint successes. Among 299 Seen and 154 Unseen jointly repaired inputs, 278 and 134 have defined source-to-repair TED for both methods. Answer-minus-Progress mean TED is 10.08 [7.53, 12.99] and 6.83 [4.58, 9.39]. Token retention improves by 5.6 [4.29, 6.94] and 6.5 [4.26, 8.95] points; line retention improves by 8.9 [6.64, 11.16] and 10.4 [6.95, 14.25]. The 21 and 20 undefined-TED cases remain in RR and are not declared unrepairable. These measures consistently describe successful outputs that retain more of the submission, but they neither establish easier comprehension nor explain which method would best help a particular student.

TABLE V
JAVA MATCHED-TARGET OUTCOMES RESOLVED BY EXERCISE AND BY STUDENT. THE LEFT COLUMNS COUNT THE 17 EXERCISES BY WHICH RULE REPAIRS MORE TRAJECTORIES IN THAT EXERCISE, WITH THE EXTRA ANSWER REPAIRS AND THE LARGEST SINGLE-EXERCISE GAP. THE RIGHT COLUMNS DESCRIBE LEXICAL-TOKEN RETENTION AMONG THE 90 TRAJECTORIES BOTH RULES REPAIR, AS THE SHARE OF THOSE STUDENTS EACH RULE FAVORS, THE MEDIAN GAIN WHERE PROGRESS IS HIGHER, AND THE MINIMUM AND MAXIMUM DIFFERENCE IN PERCENTAGE POINTS.

Repairs (17 exercises)		Retention (90 students)	
Answer higher	15	Answer higher	5.6%
Tied	2	Tied	63.3%
Progress higher	0	Progress higher	31.1%
Extra Answer repairs	63	Median gain	9.6
Largest gap	12	Retention range	-21.1 / 49.7

```
N=int(input())
S=input()
ans=0
i=0
j=N-1
while i<=j:
  while i<N:
    ...
  while j>0:
    ...
print(ans)
```

(a) submission

```
N=int(input())
S=input()
cnt=S.count('R')
ans=S[:cnt].count('W')
print(ans)
```

(b) Answer-target repair

Fig. 3. A jointly repaired Python case, selected post hoc for the largest Answer-minus-Progress TED gap and excluded from every estimate. Submission (a) passes 93.3% of tests after the student's own preceding edit changed `i<j` to `i<=j`. The Progress-target repair differs from (a) only in `j>0` becoming `j>=0` (TED 1), with the elided lines identical; the Answer-target repair (b) replaces the approach (TED 83). Both pass every test under the same gate and budget.

Java retraining uses 600 training and 118 validation pairs per target and reproduces this direction without Python AST distance. Answer targets repair 155 trajectories and Progress targets 92, a difference of 34.8 points; Progress-minus-Answer intervals are [-44.91, -25.47] under student clustering and [-47.31, -23.79] under exercise clustering. Among 90 joint repairs, Progress targets increase lexical-token retention by 4.8 points and line retention by 5.1 points, with intervals [2.79, 7.19] and [2.92, 7.25]. This is evidence for a coverage–locality tradeoff across these two languages and populations. It is not evidence that the same magnitude generalizes to other classrooms, nor a replication of the Python peer-target intervention. **What a student receives.** Cohort means do not describe an individual student, so Table V resolves the Java contrast by exercise and by student. The coverage cost is concentrated rather than uniform. One exercise accounts for 12 of the 63 additional Answer repairs, and there Progress targets repair 3 of 15 trajectories against 15. The retention gain is a minority effect with a long tail. Most students who receive a working repair from either rule get back the same amount of their own code, about a third get more, by a median of 9.6 points, and a few get less. Line retention behaves the same, zero for 64.4% of students with maximum 47.6. Figure 3 illustrates that tail on Python, where the Progress-target repair continues the student's own boundary edit while the Answer-target repair replaces the algorithm. Which rule a course deploys therefore depends on the exercise and on whether coverage or locality binds.

Peer targets. The separate outcome- and distance-matched replacement raises RR from 50.4% to 54.7% on Seen and from 78.4% to 84.4% on Unseen. Same-minus-peer intervals are [-7.71, -0.84] and [-11.79, -0.41], with exact McNemar $p = 0.005$ and $p = 0.032$. Matches cover 5,231/21,416 canonical Progress training records (24.4%) and 306/2,685 validation records (11.4%). The result concerns this caliper, donor cap, source population, and replacement rule. Equal outcomes and nearby edit distance still permit differences in target structure, style, and length, so the contrast does not isolate user identity. Matched peer targets improve coverage here; one's own trajectory is not uniquely necessary under this control.

Operational locality. A declared structural budget gives patch size an explicit engineering meaning: the deployment admits only changes within it. Per-budget Seen rates are 16.9/22.5/4.8% at TED 5, 28.5/32.4/8.0 at 10, 40.3/44.0/13.9 at 20, 51.4/50.9/30.4 at 40, 61.0/56.0/50.8 at 80, and 66.8/57.9/67.6 at 160 for current-only Answer-3Seed/Mixed/LSGen, each over 997 inputs. At TED 20, Mixed repairs 44.0% of Seen inputs versus LSGen's 13.9%, a 30.1-point difference [26.45, 33.82]. Across the six prespecified budgets its mean advantage over LSGen is 14.7 points [12.03, 17.43]. This is a budget-indexed outcome, not an unrestricted superiority claim: LSGen wins at 160, and current-only Answer-3Seed overtakes Mixed from 40 onward. On Unseen, Mixed versus current-only Answer-3Seed is 38.8% versus 32.0% at 5, 48.0% versus 40.8% at 10, and 57.6% versus

54.8% at 20; the order reverses at 40. No observed budget is retroactively designated the unique primary endpoint.

Answer to RQ2. Target construction changes coverage and locality in opposite directions under matched inputs, with the same direction on student-held-out Java, and neither effect is uniform across exercises or students. Matched peer targets improve coverage, so the effect is not uniquely personal supervision; pedagogical benefit remains unmeasured.

D. RQ3: Component Efficacy

Fixed candidate breadth. For three stored draws with success indicators a_{ij}, the expected uniformly selected one-draw success for input i is $\frac{1}{3}\sum_j a_{ij}$, whereas the three-draw union succeeds when $\max_j a_{ij} = 1$. Their paired difference isolates additional opportunities within the identical candidate set. Replacing greedy decoding by one stochastic draw costs 7.6 points [-9.81, -5.41], whereas execution-selecting three of those identical fixed draws contributes 15.1 points [13.48, 16.76] on Seen. This is a descriptive decomposition of stored executions, not a claim that three future draws have guaranteed success or independent errors. It directly demonstrates why comparing a portfolio only with one adapter overstates evidence for its specialized training signal.

The temperature sweep retains all declared conditions instead of selecting the best one for the headline. Through $T = 1.0$ the one-draw rate stays within 41.1–43.1% on Seen, the three-draw union reaches 60.0/75.2%, and at $T = 1.5$ the decoder collapses to 15.8/35.2%. A separate 20-draw stream at $T = 0.8$ reaches 74.8% Seen and 84.0% Unseen, using on average 7.44 and 5.08 calls under early stopping. Those higher rates spend a different generation budget and do not replace the three-call primary results. Fine-tuning also matters: the matched base-model three-draw union reaches only 30.4/65.6%, so adaptation adds 27.2 points [22.99, 31.41] on Seen and 6.4 points [1.19, 11.34] on Unseen. Breadth alone therefore does not explain all improvement over the base model.

At matched temperature, replacing three draws of one checkpoint by one draw from each of three checkpoints changes RR by -0.1 points [-2.00, 1.80] on Seen and 2.4 points [-0.41, 5.31] on Unseen. These intervals do not establish a consistent advantage from checkpoint diversity. They also do not show equivalence: the uncertainty still admits benefits and harms of practical interest. A deterministic multi-seed gain can coexist with this result because sampling from one checkpoint already supplies diversity.

A current-only factorial on one inference stack tests this explanation. Three greedy checkpoints gain 9.8 [7.85, 11.92] and 10.8 [7.23, 14.64] points over one, but under the matched stochastic decoder distinct checkpoints add only 2.1 points [0.00, 4.16] and 1.2 points [-2.93, 5.93]. The equal-problem Seen interval includes zero, so the rounded instance-weighted lower bound does not establish a robust checkpoint benefit. Absolute Seen/Unseen rates on that stack are 53.1/67.2% for one greedy checkpoint, 62.9/78.0% for three greedy checkpoints, 66.8/79.6% for a same-checkpoint stochastic triple, and 68.9/80.8% for a distinct-checkpoint triple, against 72.0/81.2% on the historical one. Backend sensitivity is material, and the deployment recommendation is restricted to the evaluated stack rather than asserted as backend-independent superiority.

Serialized history. Removing earlier submissions from the frozen triples raises repair for both pools. The gains are 8.8/11.1 Seen points and 7.6/6.0 Unseen points for Mixed/Answer, with all intervals excluding zero. Matched retraining without that context shows no significant difference after Holm correction, with smallest adjusted $p = 0.313$, so serialized history is not part of the measured advantage and the deployed configuration uses current-only prompts.

Selection-space controls also fail to establish a general heterogeneous-target benefit. Answer-9Choose3 receives the same nine-checkpoint pool size and 84-subset opportunity as Mixed. The primary five-fold cross-fitted mixed-minus-Answer contrast is +2.2 points [-0.19, 4.63], compared with the non-cross-fitted +1.7 [-0.79, 4.29]. These estimates are positive but uncertain. Exact replay of the six orders of a fixed triple checks the independent-candidate controller, not the value of learned diversity.

Answer to RQ3. Candidate breadth and adaptation each contribute measured gains. Distinct checkpoints add no consistent advantage once the stochastic decoder is matched, and serialized history is not supported as a necessary input. Target heterogeneity has a small, uncertain unrestricted contrast against the equally searched Answer pool.

E. RQ4: Generality

Problems and execution suites. Within Unseen, Mixed exceeds Zero-shot by 12.8 points [6.83, 18.75], with 45 versus 13 exclusive successes and exact McNemar $p = 3.01 \times 10^{-5}$. Among 142 joint repairs, 122 have defined TED for both methods; on those 122 it reduces TED by 6.41 [3.43, 9.89]. The higher absolute Unseen than Seen RR is not evidence of stronger generalization: the partitions differ in trajectory availability and problem composition. Matching all 54 Unseen problems to distinct Seen problems on six pre-repair difficulty covariates changes the raw gap into -4.6 equal-problem points [-14.18, 5.30]. We therefore rely on within-split paired contrasts, not the direction of the raw cross-split difference. Matching reduces measured imbalance but cannot balance unobserved difficulty.

A complementary observed/hidden-slice check partitions the existing authored suite. On Unseen, Mixed and Answer-9Choose3 jointly pass observed and held-back slices on 189/250 and 184/250 inputs, respectively. Conditional confirmation among observed-suite successes is 189/192 (98.4%) and 184/186 (98.9%); 13 versus 8 exclusive joint successes yield $p = 0.383$. Repeating the procedure on all 328 Seen problems with a distinct seed yields 97.8% and 97.9% confirmation, but joint success is 452/997 versus 469/997. High confirmation thus does not demonstrate that Mixed wins. These are held-back slices of the same authored tests, not independently authored specifications, and the large conditional percentage must not replace the full-denominator repair rate.

Earlier prefixes. Across 2,770 Seen and 674 Unseen executable prefixes of eventually successful trajectories, current-only Answer-3Seed reaches 69.6% and 80.1% RR. The early, middle, and last thirds reach 70.0%, 67.6%, and 71.4% on Seen and 82.9%, 77.3%, and 81.8% on Unseen. This extends evaluation beyond the last failed submission and finds no simple monotonic position effect. It does not cover trajectories that never reach acceptance, and multiple prefixes from one trajectory are dependent. The 2,770-prefix rate is neither added to the 997-input result nor reported as an independent replication on new users.

Model scale and classroom data. At 1.5B parameters three checkpoints still gain 13.4 points over one [11.16, 15.92], but mixed-minus-Answer is -0.8 [-2.62, 1.07]. Mixed reaches 50.25% and the nine-checkpoint Answer control 51.05% on Seen. This smaller model reproduces the utility of multiple opportunities, not an unrestricted mixed-target advantage. It is one scale change within a model family, not broad model-family coverage. No proprietary frontier model comparison was completed.

The Java portfolio results differ from RQ2's direct matched-target retraining. Mixed-minus-Answer-9Choose3 is positive on four held-out exercises (3.9 points [2.99, 5.11]) and inconclusive on held-out students (0.6 [-1.20, 2.59]). Absolute RR is 56.25% versus 52.30% over 304 exercise-held-out trajectories and 83.98% versus 83.43% over 181 student-held-out trajectories. The exercise estimate rests on four clusters only: refitting with one exercise left out keeps the contrast between 3.6 and 4.4 points, and single exercises range from 2.5 to 5.5 points. This sensitivity check preserves the sign on those exercises, but four clusters cannot support a broad classroom-population claim. The matched-target tradeoff has clearer cross-population support than a universal portfolio benefit.

Representation and overlap. Binary accepted/non-accepted mining removes the fine severity ordering but also changes the target set. Retraining changes Progress RR by +2.3 Seen and -2.8 Unseen points in the canonical-minus-alternative direction. The nonuniform changes show that the ordering is not an innocuous implementation detail. For Mixed and Answer-9Choose3, exact matches between a returned repair and a training-time accepted program occur in 0.6% and 1.3% of selections, near-duplicates above a 0.90 token-similarity threshold in 3.8% and 5.3%, and median maximum similarity to a same-problem training target is 66.7% and 68.6%. Low exact-copy rates do not exclude structural reuse, pretraining contamination, or user-level overlap [37].

Answer to RQ4. The matched-target tradeoff replicates on authentic student-held-out Java, and candidate breadth remains useful at smaller model scale. Mixed-target unrestricted superiority does not replicate consistently. Prefix, difficulty, overlap, verdict-order, and hidden-slice checks narrow specific threats, but they do not make CodeNet a classroom study or acceptance a proof of correctness.

F. Threats to Validity

Internal validity. Runtime variability, undefined source ASTs, and Python whitespace affect labels and distances. We retain unknowns, denominators, and whitespace-sensitive corrections; hashes establish provenance, not correctness. Source and peer matching do not equate all target semantics or styles.
Construct validity. Acceptance is not proof of semantic correctness [38]; same-suite slices are not independent specifications. TED and retention do not measure cognitive load or subsequent learning. CodeNet does not identify novices or measure mastery. Java supplies classroom submissions, not an intervention observing feedback use. The per-exercise and per-student resolution describes which students would receive a different repair, not which students benefit, and Figure 3 is one post-hoc case that carries no estimate. We display a CodeNet program because the course data terms do not permit redistributing student submissions. These distinctions limit educational and target-quality interpretations.
External and conclusion validity. Filtering favors short, eventually successful Python trajectories and permits Seen user overlap. Java adds one course context and only four held-out exercises. Cluster bootstrap and equal-problem estimates do not eliminate small-cluster uncertainty, multiplicity, or the bias of analyses chosen after earlier results.

V. Related Work

Repair and generation budgets. Generate-and-validate repair spans search, learned transformations, and language models [1]–[4]. Search analyses [39] and execution-based evaluation [40], [41] separate candidate quality from exposure. Neural patches, ensembles, synthetic corruption, and execution-informed self-supervision provide different training signals [26], [42]–[44]; zero-shot studies examine repair without task adaptation [45], [46]. Independent ensembles and weight averaging supply other notions of diversity [47], [48]. We do not introduce portfolio repair; we isolate trajectory targets with same-target seeds, equal search spaces, fixed draws, and current-only controls.
Programming-exercise feedback. Educational repair exploits correct programs through clustering, alignment, and refactoring [10]–[12]; related representations and tutoring systems propagate feedback and generate hints [17]–[19], [49]. Language models support repair, syntax feedback, conversations, peers, and counterexamples [6], [7], [13]–[15], [50], [51]. Assignment-aware and diversified feedback use further reference strategies [52]–[54]. LSGen also mines chronological wrong-to-later-correct submission pairs for retrieval [16], so trajectories are not our novelty, and it retains its strong Seen advantage here. Next-step hint generators differ from this work in what their guidance is fitted to. Data-driven hints read a solution space or a past cohort's paths and project the submission onto it, so two students at the same position in that space receive the same next step independently of which tests their own code currently passes [18]–[20]. Our target rule reads the submission's own execution evidence instead, and returns a complete program rather than a textual hint.

The two kinds of output are measured on different scales and do not substitute for each other. Benchmarking and surveys emphasize population and protocol differences [5], [21], [55]. These methods do not uniformly train on Answer targets. Our narrower intervention replaces the target for matched sources and exposes a coverage–locality tradeoff, not dominance over feedback strategies.

History and learner evidence. ZPD and scaffolding concern supported performance, not program edit distance [22]–[25]. Knowledge tracing models proficiency from interactions [56]–[58]; TIKTOC incorporates testcase evidence [30]. Next-step hints, feedback ladders, and novice-response studies examine guidance more directly [20], [59], [60], while feedback theory emphasizes its interpretation and use [8], [9], [61], [62]. This paper does not claim that conditioning on a learner's serialized history improves generated repairs. Our deployed configuration prompts with the current submission alone, and the supplement reports the matched retraining and prompt controls behind that choice, including their negative outcomes. Educational claims additionally require comprehension, transfer, and independent-correction measurements, not only executable outputs.

VI. Conclusion

We presented ZPDPatch, a Mine–Train–Select technique that constructs repair targets from a submission's own execution evidence instead of from an accepted solution. Matched Python and student-held-out Java experiments show the same coverage–locality tradeoff. Progress targets preserve more of the student's source on joint successes while repairing fewer inputs, and on part of what they fail to repair they still return a program that has moved forward. Candidate breadth and model adaptation account for the measurable part of the remaining improvement, and matched peer targets also raise coverage, so the retained-evidence target is not uniquely personal supervision. Resolved per exercise and per student, that tradeoff is uneven. The coverage cost concentrates on particular exercises, and the retention gain reaches about a third of the students who receive a repair from either rule, which is where a course would have to choose. We therefore retain the current-only Answer triple and report the Progress rule as the option for when locality binds. Educational benefit and broader model-family generality remain unmeasured.

Data Availability

An anonymous supplement contains code-free paired outcomes, condition manifests, metric code, negative results, and claim–evidence links. Its offline verifier checks hashes and recomputes matched Python/Java contrasts. Student source and model weights are excluded: the CodeWorkout/TIKTOC terms govern redistribution of student submissions [30], CodeNet retains its own terms [29], and the adapters exceed the supplement budget. We therefore share the derived outcome and condition data that recompute every reported contrast, as anonymous supplementary material rather than a public link, which keeps review double-blind. The supplement also carries the complete inventory of the development runs that this paper does not report, including their negative outcomes. No student intervention was conducted.

References

[1] L. Gazzola, D. Micucci, and L. Mariani, "Automatic software repair: A survey," *IEEE Transactions on Software Engineering*, vol. 45, no. 1, pp. 34–67, 2019.

[2] M. Monperrus, "Automatic software repair: A bibliography," *ACM Computing Surveys*, vol. 51, no. 1, pp. 1–24, 2018.

[3] W. Weimer, T. Nguyen, C. Le Goues, and S. Forrest, "Automatically finding patches using genetic programming," in *Proceedings of the 31st International Conference on Software Engineering*, 2009, pp. 364–374.

[4] C. Le Goues, T. Nguyen, S. Forrest, and W. Weimer, "GenProg: A generic method for automatic software repair," *IEEE Transactions on Software Engineering*, vol. 38, no. 1, pp. 54–72, 2012.

[5] H. Keuning, J. Jeuring, and B. Heeren, "A systematic literature review of automated feedback generation for programming exercises," *ACM Transactions on Computing Education (TOCE)*, vol. 19, no. 1, pp. 1–43, 2018.

[6] H. Joshi, J. C. Sanchez, S. Gulwani, V. Le, G. Verbruggen, and I. Radiček, "Repair is nearly generation: Multilingual program repair with LLMs," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 4, 2023, pp. 5131–5140.

[7] J. Zhang, J. P. Cambronero, S. Gulwani, V. Le, R. Piskac, G. Soares, and G. Verbruggen, "PyDex: Repairing bugs in introductory Python assignments using LLMs," *Proceedings of the ACM on Programming Languages*, vol. 8, no. OOPSLA1, pp. 1100–1124, 2024.

[8] S. Narciss, "Feedback strategies for interactive learning tasks," in *Handbook of research on educational communications and technology*. Routledge, 2008, pp. 125–143.

[9] V. J. Shute, "Focus on formative feedback," *Review of Educational Research*, vol. 78, no. 1, pp. 153–189, 2008.

[10] S. Gulwani, I. Radiček, and F. Zuleger, "Automated clustering and program repair for introductory programming assignments," *ACM SIGPLAN Notices*, vol. 53, no. 4, pp. 465–480, 2018.

[11] K. Wang, R. Singh, and Z. Su, "Search, align, and repair: data-driven feedback generation for introductory programming exercises," in *Proceedings of the 39th ACM SIGPLAN conference on programming language design and implementation*, 2018, pp. 481–495.

[12] Y. Hu, U. Z. Ahmed, S. Mechtaev, B. Leong, and A. Roychoudhury, "Re-factoring based program repair applied to programming assignments," in *Proceedings of the 34th IEEE/ACM International Conference on Automated Software Engineering (ASE)*. IEEE, 2019, pp. 388–398.

[13] F. Liu, Z. Liu, Q. Zhao, J. Jiang, L. Zhang, Z. Sun, G. Li, Z. Li, and Y. Ma, "FastFixer: An efficient and effective approach for repairing programming assignments," in *Proceedings of the 39th IEEE/ACM International Conference on Automated Software Engineering*, 2024, pp. 669–680.

[14] B. Yang, H. Tian, W. Pian, H. Yu, H. Wang, J. Klein, T. F. Bissyandé, and S. Jin, "CREF: An LLM-based conversational software repair framework for programming tutors," in *Proceedings of the 33rd ACM SIGSOFT International Symposium on Software Testing and Analysis*, 2024, pp. 882–894.

[15] P. Orvalho, M. Janota, and V. M. Manquinho, "Counterexample guided program repair using zero-shot learning and maxsat-based fault localization," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 1, 2025, pp. 649–657.

[16] Z. Dai, Z. Zhao, H. Wang, X. Tang, S. Wu, C. Yao, Z. Gao, and J. Chen, "Learner-tailored program repair: A solution generator with iterative edit-driven retrieval enhancement," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 36, 2026, pp. 30 467–30 475.

[17] C. Piech, J. Huang, A. Nguyen, M. Phulsuksombati, M. Sahami, and L. Guibas, "Learning program embeddings to propagate feedback on student code," in *Proceedings of the 32nd International Conference on Machine Learning*, 2015, pp. 1093–1102.

[18] K. Rivers and K. R. Koedinger, "Data-driven hint generation in vast solution spaces: A self-improving python programming tutor," *International Journal of Artificial Intelligence in Education*, vol. 27, no. 1, pp. 37–64, 2017.

[19] T. W. Price, Y. Dong, and D. Lipovac, "iSnap: towards intelligent tutoring in novice programming environments," in *Proceedings of the 2017 ACM SIGCSE Technical Symposium on computer science education*, 2017, pp. 483–488.

[20] L. Roest, H. Keuning, and J. Jeuring, "Next-step hint generation for introductory programming using large language models," in *Proceedings of the 26th Australasian Computing Education Conference*, 2024, pp. 144–153.

[21] C. Koutcheme, N. Dainese, S. Sarsa, J. Leinonen, A. Hellas, and P. Denny, "Benchmarking educational program repair," *arXiv preprint arXiv:2405.05347*, 2024.

[22] L. S. Vygotsky, *Mind in Society: The Development of Higher Psychological Processes*, M. Cole, V. John-Steiner, S. Scribner, and E. Souberman, Eds. Cambridge, MA: Harvard University Press, 1978.

[23] T. Murray and I. Arroyo, "Toward measuring and maintaining the zone of proximal development in adaptive instructional systems," in *International conference on intelligent tutoring systems*. Springer, 2002, pp. 749–758.

[24] I.-A. Chounta, P. Albacete, P. Jordan, S. Katz, and B. M. McLaren, "The "grey area": A computational approach to model the zone of proximal development," in *European conference on technology enhanced learning*. Springer, 2017, pp. 3–16.

[25] N. Schulze Bernd and I.-A. Chounta, "Beyond the grey area: exploring the effectiveness of scaffolding as a learning measure," in *International Conference on Artificial Intelligence in Education*. Springer, 2024, pp. 365–378.

[26] M. Tufano, C. Watson, G. Bavota, M. Di Penta, M. White, and D. Poshyvanyk, "An empirical study on learning bug-fixing patches in the wild via neural machine translation," *ACM Transactions on Software Engineering and Methodology*, vol. 28, no. 4, pp. 1–29, 2019.

[27] B. Hui, J. Yang, Z. Cui, J. Yang, D. Liu, L. Zhang, T. Liu, J. Zhang, B. Yu, K. Lu *et al.*, "Qwen2.5-Coder technical report," *arXiv preprint arXiv:2409.12186*, 2024.

[28] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," *Advances in neural information processing systems*, vol. 36, pp. 10 088–10 115, 2023.

[29] R. Puri, D. Kung, G. Janssen, W. Zhang, G. Domeniconi, V. Zolotov, J. Dolby, J. Chen, M. Choudhury, L. Decker *et al.*, "CodeNet: A large-scale AI for code dataset for learning a diversity of coding tasks," in *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, vol. 1, 2021.

[30] Z. Duan, N. Fernandez, A. Hicks, and A. Lan, "Test case-informed knowledge tracing for open-ended coding tasks," in *Proceedings of the 15th International Learning Analytics and Knowledge Conference*, 2025, pp. 238–248.

[31] K. Zhang and D. Shasha, "Simple fast algorithms for the editing distance between trees and related problems," *SIAM Journal on Computing*, vol. 18, no. 6, pp. 1245–1262, 1989.

[32] M. Pawlik and N. Augsten, "Tree edit distance: Robust and memory-efficient," *Information Systems*, vol. 56, pp. 157–173, 2016.

[33] B. Efron, "Bootstrap methods: Another look at the jackknife," *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979.

[34] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.

[35] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.

[36] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927.

[37] M. Allamanis, "The adverse effects of code duplication in machine learning models of code," in *Proceedings of the 2019 ACM SIGPLAN International Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software*, 2019, pp. 143–153.

[38] E. K. Smith, E. T. Barr, C. Le Goues, and Y. Brun, "Is the cure worse than the disease? overfitting in automated program repair," in *Proceedings of the 10th Joint Meeting on Foundations of Software Engineering*, 2015, pp. 532–543.

[39] Y. Qi, X. Mao, Y. Lei, Z. Dai, and C. Wang, "The strength of random search on automated program repair," in *Proceedings of the 36th International Conference on Software Engineering*, 2014, pp. 254–265.

[40] M. M. A. Haque, W. U. Ahmad, I. Lourentzou, and C. Brown, "Fix-Eval: Execution-based evaluation of program fixes for programming problems," in *2023 IEEE/ACM International Workshop on Automated Program Repair (APR)*. IEEE, 2023, pp. 11–18.

[41] M. Chen, J. Tworek, H. Jun, Q. Yuan, H. P. de Oliveira Pinto, J. Kaplan, H. Edwards, Y. Burda, N. Joseph, G. Brockman *et al.*, "Evaluating large

language models trained on code," *arXiv preprint arXiv:2107.03374*, 2021.

[42] T. Lutellier, H. V. Pham, L. Pang, Y. Li, M. Wei, and L. Tan, "Coconut: Combining context-aware neural translation models using ensemble for program repair," in *Proceedings of the 29th ACM SIGSOFT International Symposium on Software Testing and Analysis*, 2020, pp. 101–114.

[43] M. Yasunaga and P. Liang, "Break-it-fix-it: Unsupervised learning for program repair," in *International conference on machine learning*. PMLR, 2021, pp. 11 941–11 952.

[44] H. Ye, M. Martinez, X. Luo, T. Zhang, and M. Monperrus, "SelfAPR: Self-supervised program repair with test execution diagnostics," in *Proceedings of the 37th IEEE/ACM International Conference on Automated Software Engineering*, 2022, pp. 1–13.

[45] C. S. Xia and L. Zhang, "Less training, more repairing please: Revisiting automated program repair via zero-shot learning," in *Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, 2022, pp. 959–971.

[46] N. Jiang, K. Liu, T. Lutellier, and L. Tan, "Impact of code language models on automated program repair," in *Proceedings of the 45th International Conference on Software Engineering*, 2023, pp. 1430–1442.

[47] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[48] M. Wortsman, G. Ilharco, S. Y. Gadre, R. Roelofs, R. Gontijo-Lopes, A. S. Morcos, H. Namkoong, A. Farhadi, Y. Carmon, S. Kornblith, and L. Schmidt, "Model soups: Averaging weights of multiple fine-tuned models improves accuracy without increasing inference time," in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162, 2022, pp. 23 965–23 998.

[49] E. L. Glassman, J. Scott, R. Singh, P. J. Guo, and R. C. Miller, "Overcode: Visualizing variation in student solutions to programming problems at scale," *ACM Transactions on Computer-Human Interaction*, vol. 22, no. 2, pp. 1–35, 2015.

[50] T. Phung, J. Cambronero, S. Gulwani, T. Kohn, R. Majumdar, A. Singla, and G. Soares, "Generating high-precision feedback for programming syntax errors using large language models," in *Proceedings of the 16th International Conference on Educational Data Mining*, 2023, pp. 370–377.

[51] Q. Zhao, F. Liu, L. Zhang, Y. Liu, Z. Yan, Z. Chen, Y. Zhou, J. Jiang, and G. Li, "Peer-aided repairer: Empowering large language models to repair advanced student assignments," *arXiv preprint arXiv:2404.01754*, 2024.

[52] J. Heo, H. Jeong, D. Choi, and E. Lee, "REFERENT: Transformer-based feedback generation using assignment information for programming course," in *2023 IEEE/ACM 45th International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)*. IEEE, 2023, pp. 101–106.

[53] D. Choi, J. Heo, and E. Lee, "Automated feedback generation for multiple function programs," in *2021 28th Asia-Pacific Software Engineering Conference (APSEC)*. IEEE, 2021, pp. 582–583.

[54] D. Choi and E. Lee, "Automated feedback generation for programming assignments through diversification," in *2025 IEEE/ACM 37th International Conference on Software Engineering Education and Training (CSEE&T)*. IEEE, 2025, pp. 230–241.

[55] J. Yi, U. Z. Ahmed, A. Karkare, S. H. Tan, and A. Roychoudhury, "A feasibility study of using automated program repair for introductory programming assignments," in *Proceedings of the 2017 11th joint meeting on foundations of software engineering*, 2017, pp. 740–751.

[56] A. T. Corbett and J. R. Anderson, "Knowledge tracing: Modeling the acquisition of procedural knowledge," *User modeling and user-adapted interaction*, vol. 4, no. 4, pp. 253–278, 1994.

[57] C. Piech, J. Bassen, J. Huang, S. Ganguli, M. Sahami, L. J. Guibas, and J. Sohl-Dickstein, "Deep knowledge tracing," *Advances in neural information processing systems*, vol. 28, 2015.

[58] A. Ghosh, N. Heffernan, and A. S. Lan, "Context-aware attentive knowledge tracing," in *Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining*, 2020, pp. 2330–2339.

[59] H. Heickal and A. Lan, "Generating feedback-ladders for logical errors in programming using large language models," in *Proceedings of the 17th International Conference on Educational Data Mining*, 2024, pp. 947–951.

[60] R. Xiao, X. Hou, and J. Stamper, "Exploring how multiple levels of gpt-generated programming hints support or disappoint novices," in *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, 2024, pp. 1–10.

[61] J. Hattie and H. Timperley, "The power of feedback," *Review of Educational Research*, vol. 77, no. 1, pp. 81–112, 2007.

[62] V. Aleven and K. R. Koedinger, "Limitations of student control: Do students know when they need help?" in *International conference on intelligent tutoring systems*. Springer, 2000, pp. 292–303.